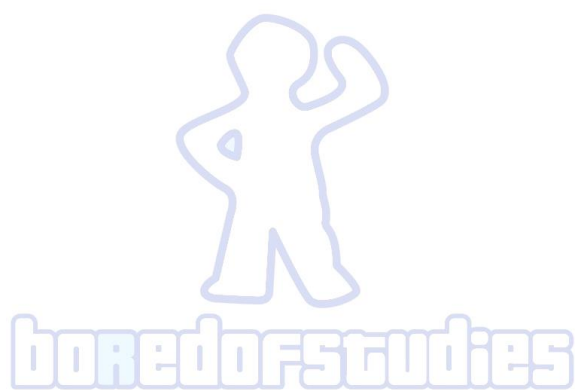




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2020

BORED OF STUDIES TRIAL EXAMINATION

9th October

Mathematics Extension 2

General instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using a black or blue pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

Total marks: 100**Section I – 10 marks** (pages 2–4)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 5–14)

- Attempt Questions 11–16
- Allow about 2 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 Consider the vector equations of the lines below.

$$\vec{r} = \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 4 \\ -3 \end{pmatrix} \quad \text{and} \quad \vec{r} = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ 8 \\ 1 \end{pmatrix}.$$

Which of the following statements is correct?

- (A) The lines are parallel with no point of intersection.
- (B) The lines are not parallel with no point of intersection.
- (C) The lines are parallel with a point of intersection.
- (D) The lines are not parallel with a point of intersection.

- 2 Consider the region described by the following inequalities over the complex plane.

$$z + \bar{z} \leq 0 \quad \text{and} \quad -\frac{\pi}{6} \leq \arg(z + 1 - i) \leq \frac{\pi}{6}.$$

What is the area of the region?

- | | |
|--------------------------|--------------------------|
| (A) $\frac{2}{\sqrt{3}}$ | (B) $\frac{1}{\sqrt{3}}$ |
| (C) $\sqrt{3}$ | (D) $\frac{\sqrt{3}}{2}$ |

- 3 Let $f(x)$ and $g(x)$ be real-valued functions, over the domain $x \geq 0$.

Suppose that $f(x) \geq 0$, $\lim_{x \rightarrow 0^+} f(x) = 0$ and $\lim_{x \rightarrow 0^+} g(x) = 0$.

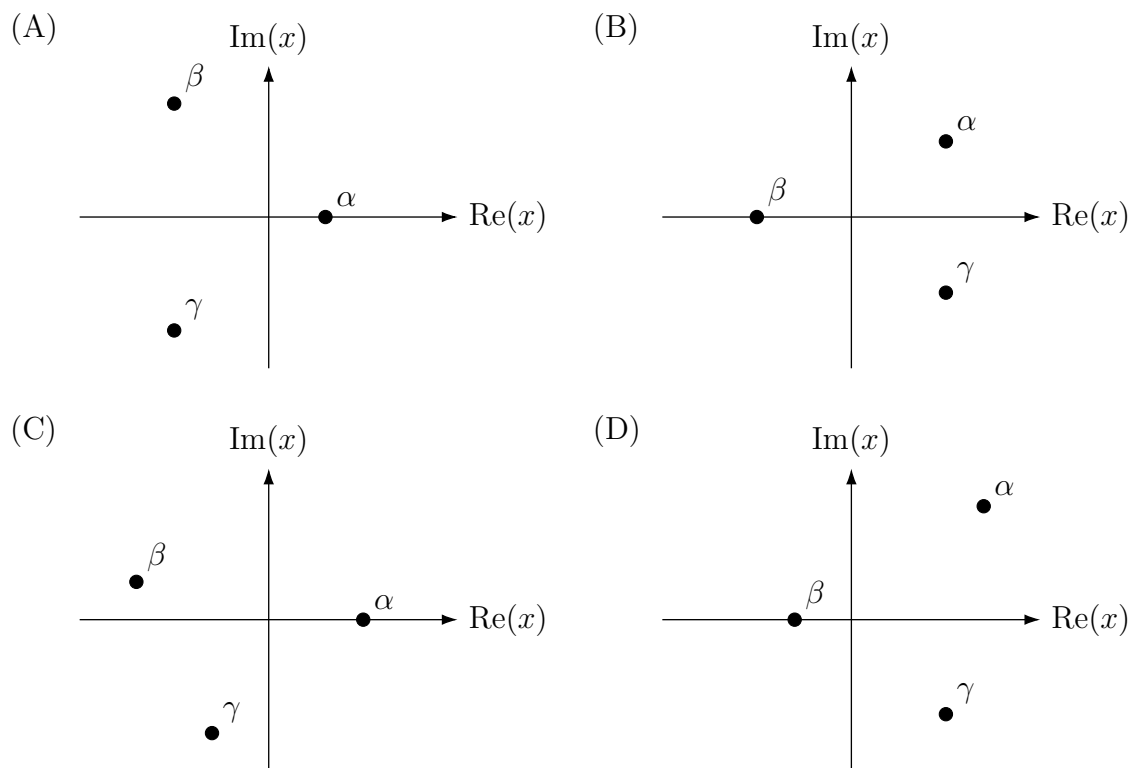
Further suppose that $\lim_{x \rightarrow 0^+} (f(x))^{g(x)}$ exists, and is equal to L .

Which statement best describes the value of L ?

- | | |
|-------------|--|
| (A) $L = 0$ | (B) L can be any non-negative real number. |
| (C) $L = 1$ | (D) L can be any real number. |

- 4 Consider the polynomial $P(x) = x^3 + ax^2 - bx + c$ where a, b and c are positive constants. The graph of $y = P(x)$ has the stationary points (x_1, y_1) and (x_2, y_2) in the first and second quadrant respectively.

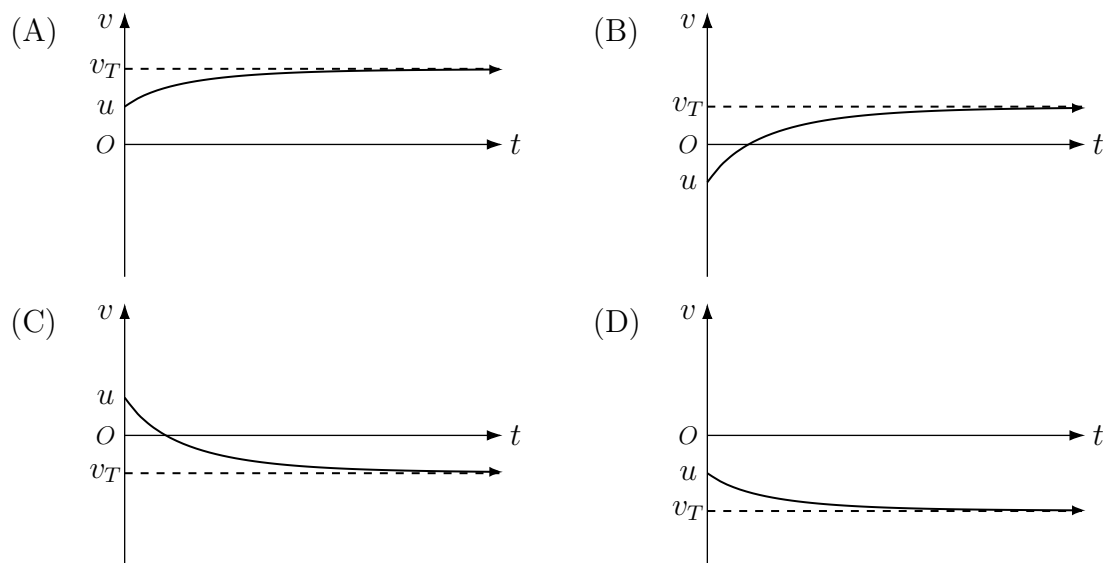
Which diagram could represent the location of the roots of $P(x)$ in the complex plane?



- 5 An object is projected directly upwards with an initial velocity of u . It experiences a fixed gravitational force and air resistance during its trajectory. The object eventually reaches a terminal velocity of v_T .

Let upwards be the positive direction.

Which of the following graphs best describes the object's velocity over time?



- 6 The vector equation of a line in three dimensional space for some scalar λ is given by

$$\underline{r} = \underline{a} + \lambda \hat{n}$$

where $\underline{a}$ is a position vector and $\hat{n}$ is a *unit* vector in the direction of the line.

Let $\underline{p}$ be some point which does not lie on the line. Which of the following represents the shortest distance from $\underline{p}$ to the line?

- (A) $|((\underline{a} - \underline{p}) \cdot \hat{n})\hat{n}|$ (B) $|((\underline{a} - \underline{p}) \cdot \hat{n})\hat{n} - \hat{n}|$
 (C) $|((\underline{a} - \underline{p}) \cdot \hat{n})\hat{n} - (\underline{a} - \underline{p})|$ (D) $|((\underline{a} - \underline{p}) \cdot \hat{n})\hat{n} - \underline{p}|$

- 7 A particle is moving in simple harmonic motion according to the acceleration equation $\ddot{x} = -kx$ where k is a known positive constant and x is the particle's displacement at time t . Given that the particle is initially at the origin, and the information described above, which of the following *cannot* be determined?

- (A) Period (B) Amplitude
 (C) Centre of motion (D) Initial acceleration

- 8 Let $f(x)$ be a function over some domain D . Which of the following statements could imply that $\lim_{x \rightarrow 0} f(x) = 0$?

- (A) $\forall \epsilon > 0, \exists \delta > 0$ such that $\forall x \in D, |x| < \delta \Rightarrow |f(x)| < \epsilon$
 (B) $\forall \epsilon > 0, \exists \delta > 0$ such that $\forall x \in D, |x| < \delta \Rightarrow |f(x)| > \epsilon$
 (C) $\forall \epsilon > 0, \exists \delta > 0$ such that $\forall x \in D, |x| > \delta \Rightarrow |f(x)| < \epsilon$
 (D) $\forall \epsilon > 0, \exists \delta > 0$ such that $\forall x \in D, |x| > \delta \Rightarrow |f(x)| > \epsilon$

- 9 Let A, B and C be non-collinear points represented by complex numbers a, b and c respectively on the complex plane. Which condition implies $\triangle ABC$ is isosceles?

- (A) $|b|^2 + a\bar{c} + \bar{a}c = |c|^2 + a\bar{b} + \bar{a}b$ (B) $|b|^2 + a\bar{b} + \bar{a}b + a\bar{c} + \bar{a}c = |c|^2$
 (C) $|b|^2 = |c|^2 + a\bar{c} + \bar{a}c + a\bar{b} + \bar{a}b$ (D) $|b|^2 + a\bar{b} + \bar{a}b = |c|^2 + a\bar{c} + \bar{a}c$

- 10 Which of the following is equivalent to $\int_0^1 \frac{1-x^2}{1+x^2} dx$?

- (A) $\int_0^{\frac{\pi}{4}} \frac{\tan x}{\tan 2x} dx$ (B) $\int_0^1 x \tan^{-1} x dx$
 (C) $\int_0^{\frac{\pi}{4}} \frac{x \sin x}{\cos^3 x} dx$ (D) $\int_0^1 \left(\frac{1}{2} \ln(1+x^2) + \tan^{-1} x \right) dx$

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the Question 11 Writing Booklet

- (a) Use the substitution $x = \frac{1}{u}$ to evaluate 4

$$\int_0^{\infty} \frac{1}{x^8 + x^4 + 1} dx.$$

You may assume, without proof, that

$$a^3 + 1 = (a + 1)(a^2 - a + 1).$$

- (b) Consider the quadratic polynomial $P(x) = ax^2 + bx + c$ where a , b and c are real. 2
 $P(x)$ has non-real solutions z_1 and z_2 , which lie on the complex plane.

Find the equation of the circle with the smallest possible radius passing through both z_1 and z_2 in terms of a , b and c .

- (c) Let $\underline{a}_1, \underline{a}_2, \underline{b}_1$ and $\underline{b}_2$ be arbitrary fixed position vectors in three dimensional space. 2
Suppose that two distinct lines ℓ_1 and ℓ_2 are represented by the vector equations

$$\ell_1 : \underline{r} = \underline{a}_1 + \lambda_1 \underline{b}_1$$

$$\ell_2 : \underline{r} = \underline{a}_2 + \lambda_2 \underline{b}_2$$

where λ_1 and λ_2 are scalar constants.

Let $\underline{n}$ be a vector that is simultaneously perpendicular to both $\underline{b}_1$ and $\underline{b}_2$.
If the lines ℓ_1 and ℓ_2 have a point of intersection, show that

$$(\underline{a}_1 - \underline{a}_2) \cdot \underline{n} = 0.$$

Question 11 continues on page 6

Question 11 (continued)

- (d) An object is initially at the origin and moves horizontally in the positive direction of the x -axis with an initial speed of u . It is subjected to a resistance force such that its acceleration equation for some integer $n > 2$ is

$$\ddot{x} = -kv^n$$

where v is the object's speed at time t and k is a positive constant.

- (i) Show that the displacement equation is **2**

$$x = \frac{u^{n-2} - v^{n-2}}{k(n-2)(uv)^{n-2}}.$$

- (ii) By considering the average speed of the object from the origin to a particular positive displacement x , show that **3**

$$v < \frac{a_{n-1}}{a_{n-2}} < u$$

$$\text{where } a_n = \frac{u^n - v^n}{n}.$$

- (e) Let $a_1, a_2, \dots, a_{n-1}$ and a_n be distinct non-zero real numbers. Define the polynomial

$$P(x) = (x^2 + a_1^2)(x^2 + a_2^2) \dots (x^2 + a_n^2).$$

- (i) Show that there exists some real constants $c_1, c_2, \dots, c_{n-1}$ and c_n such that **1**

$$\frac{1}{P(x)} = \frac{c_1}{x^2 + a_1^2} + \frac{c_2}{x^2 + a_2^2} + \dots + \frac{c_n}{x^2 + a_n^2}.$$

- (ii) Hence, express $c_1, c_2, \dots, c_{n-1}$ and c_n in terms of $a_1, a_2, \dots, a_{n-1}$ and a_n . **1**

End of Question 11

Question 12 (15 marks) Use the Question 12 Writing Booklet

- (a) Let $z_1, z_2, z_3, \dots, z_n$ be the complex roots of $z^n = 1$ where n is a positive integer. **2**

Suppose that $\arg(z_{i+1}) - \arg(z_i) = \frac{2\pi}{n}$ for $i = 1, 2, 3, \dots, n-1$. Show that

$$(z_1 - z_2)^2 + (z_2 - z_3)^2 + (z_3 - z_4)^2 + \dots + (z_{n-1} - z_n)^2 + (z_n - z_1)^2 = 0.$$

- (b) A particle moves on the x - y plane according to the acceleration vector

$$\underline{a} = -n^2(x\underline{i} + y\underline{j})$$

where n is a positive constant. The particle is initially at the point $(A, B \cos \theta)$ with velocity $\underline{v} = -Bn \sin \theta \underline{j}$ where A, B and θ are positive constants.

- (i) Find the displacement position vector $\underline{r}$ in terms of time t . **2**

- (ii) Hence, show that the Cartesian equation of the path of the particle is **2**

$$\frac{x^2}{A^2} - \frac{2xy \cos \theta}{AB} + \frac{y^2}{B^2} = \sin^2 \theta.$$

- (iii) Describe the path of the particle and the *direction* of the motion on the number plane for $\theta = \frac{\pi}{2}$. **2**

- (c) Let α, β and γ be the roots of the polynomial $P(x) = x^3 + bx^2 + cx + d$ where b, c and d are real, non-zero constants. It is known that **3**

- $\alpha + \beta = 2$; and
- All the roots of $P(x)$ also satisfy the equation $|x^3| = 8$.

Find the values of b, c and d .

- (d) Let $\underline{u}$ and $\underline{v}$ be two distinct vectors in three dimensions. Let a_1 and a_2 be scalars.

Define $\underline{0} = 0\underline{i} + 0\underline{j} + 0\underline{k}$ and suppose that $\underline{u} \neq \underline{0}$ and $\underline{v} \neq \underline{0}$.

Consider the following statement:

If $\underline{u} \cdot \underline{v} = 0$ then the only solution to $a_1\underline{u} + a_2\underline{v} = \underline{0}$ is $a_1 = a_2 = 0$.

- (i) Prove that the above statement is always true. **2**

- (ii) Is the converse of the statement always true? Justify your answer with either a proof or a counterexample. **2**

End of Question 12

Question 13 (15 marks) Use the Question 13 Writing Booklet

- (a) Let z_1, z_2 and z_3 be distinct complex numbers such that for non-zero real values a and b

$$z_3 = \frac{az_1 + ibz_2}{a + ib}.$$

Suppose that A, B and C represent z_1, z_2 and z_3 respectively on the complex plane.

- (i) Show that $\triangle ACB$ is a right-angled triangle. **2**

- (ii) Suppose that $\triangle ACB$ is also an isosceles right-angled triangle. **1**
State the restrictions on the values of a and b .

- (b) Suppose a, b and c are positive real numbers such that $ab + bc + ca = 3$.

- (i) Show that **1**

$$abc \leq 1.$$

You may assume, without proof, that

$$\frac{x + y + z}{3} \geq (xyz)^{\frac{1}{3}}.$$

- (ii) Hence, show that **2**

$$\frac{1}{1 + a^2(b + c)} + \frac{1}{1 + b^2(a + c)} + \frac{1}{1 + c^2(a + b)} \leq \frac{1}{abc}.$$

Question 13 continues on page 9

Question 13 (continued)

- (c) Let $H_0(x) = 1$ and define

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} (e^{-x^2})$$

where n is a positive integer and $\frac{d^n}{dx^n} (e^{-x^2})$ refers to the n th derivative of e^{-x^2} .

- (i) Prove by mathematical induction that **3**

$$H_n(x) = 2xH_{n-1}(x) - \frac{d}{dx}H_{n-1}(x).$$

- (ii) Hence, explain why $H_n(x)$ is a polynomial of degree n . **2**

- (d) Let $\underline{x}$ be a variable position vector X on a sphere with the vector equation

$$|\underline{x} - \underline{c}| = r$$

where $\underline{c}$ is the position vector of the centre C and r is the radius of the sphere.

Let $\underline{x}_0$ be a position vector of a fixed point P , lying on the surface of the sphere.

- (i) Explain why the vector equation of the plane at P , perpendicular to CP is **1**

$$(\underline{x} - \underline{x}_0) \cdot (\underline{x}_0 - \underline{c}) = 0$$

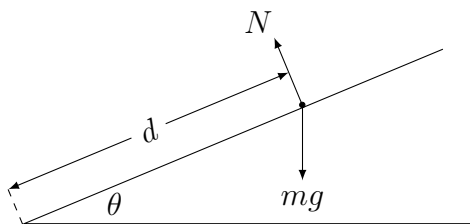
where $\underline{x}$ is a variable point on the plane.

- (ii) Hence, use the vector equations of the sphere and the plane to algebraically show that they touch only at the point P . **3**

End of Question 13

Question 14 (15 marks) Use the Question 14 Writing Booklet

- (a) An object of mass m is initially at rest on a frictionless surface at an angle of θ to the horizontal and a distance of d to the end of the slope. The two forces on the object are the normal force of magnitude N and the gravitational force of magnitude mg .



The direction of the net force on the object is parallel to the inclined surface.

- (i) Explain why $N = mg \cos \theta$. 1
- (ii) Show that the speed of the object at the base of the inclined surface is 2
- $$v_d = \sqrt{2gd \sin \theta}.$$
- (iii) Let t_d be the time taken for the object to reach the base of the inclined surface. 2
- Show that $d = \frac{v_d t_d}{2}$.

- (b) Let $z = e^{i\theta}$ where $\theta \neq 2m\pi$ for any integer m .

- (i) Show that for non-negative integers n 2

$$\sum_{k=0}^n \operatorname{Re}(z^k) = \left(\frac{\sin\left(\frac{(n+1)\theta}{2}\right)}{\sin\frac{\theta}{2}} \right) \operatorname{Re}(z^{\frac{n}{2}}).$$

- (ii) Hence, show that $\sum_{k=1}^n k \sin(k\theta) = \frac{(n+1) \sin(n\theta) - n \sin[(n+1)\theta]}{2(1 - \cos \theta)}$. 3

- (c) Four distinct points A, B, C and D in three dimensional space are represented by the vectors $\underline{a}, \underline{b}, \underline{c}$ and $\underline{d}$ respectively. Suppose that no three points are collinear and $\angle ABC + \angle ADC = \pi$.

- (i) If D lies on the same plane as A, B and C , making a quadrilateral $ABCD$, show that $|\underline{b} - \underline{a}||\underline{d} - \underline{c}| + |\underline{d} - \underline{a}||\underline{c} - \underline{b}| = |\underline{c} - \underline{a}||\underline{d} - \underline{b}|$. 4

- (ii) Suppose that D does not lie on the same plane as A, B and C . Explain why 1

$$|\underline{b} - \underline{a}||\underline{d} - \underline{c}| + |\underline{d} - \underline{a}||\underline{c} - \underline{b}| > |\underline{c} - \underline{a}||\underline{d} - \underline{b}|.$$

End of Question 14

Question 15 (15 marks) Use the Question 15 Writing Booklet

(a) Let $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx$ for non-negative integers n .

(i) Show that for $n \geq 2$

2

$$I_n = \frac{n-1}{n} I_{n-2}.$$

(ii) Hence show that

2

$$I_{2n} = \frac{\pi}{2} \times \frac{(2n)!}{2^{2n}(n!)^2}.$$

(iii) Show that $\frac{2n+1}{2n+2} \leq \frac{I_{2n+1}}{I_{2n}} \leq 1$.

2

(iv) It can be shown that

1

$$I_{2n+1} = \frac{2^{2n}(n!)^2}{(2n+1)!} \quad (\text{Do NOT prove this}).$$

Use this result and the results in part (ii) and part (iii) to deduce that

$$\lim_{n \rightarrow \infty} \frac{2^{4n}(n!)^4}{(2n)!(2n+1)!} = \frac{\pi}{2}.$$

(v) Let $u_n = \frac{e^n n!}{n^{n+\frac{1}{2}}}$ be a sequence of real numbers for non-negative integers of n .

2

It can be shown that as $n \rightarrow \infty$ then $u_n \rightarrow L$, where L is a positive constant.

Using the result in part (iv), show that $L = \sqrt{2\pi}$.

Question 15 continues on page 12

Question 15 (continued)

- (b) Show that for $x > 0$

2

$$0 < \ln(1+x) - x + \frac{x^2}{2} < \frac{x^3}{3}.$$

- (c) A random variable X which only takes positive values, based on a sample of size n , has a probability density function of

$$f(x) = \frac{x^n e^{-x}}{n!}$$

where n is a positive integer.

Let $Z = \frac{X - n}{\sqrt{n}}$ be a random variable with a probability density function $g(z)$.

- (i) Show that

1

$$g(z) = \sqrt{n} f(n + z\sqrt{n}).$$

- (ii) Hence, using the results in parts (a)(v) and (b), show that for $z > 0$

3

$$\lim_{n \rightarrow \infty} g(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}.$$

End of Question 15

Question 16 (15 marks) Use the Question 16 Writing Booklet

- (a) A student launches an object from the ground at an angle of θ to the horizontal with an initial speed of u . The object is subjected to air resistance such that the acceleration equations are

$$\begin{aligned}\ddot{x} &= -\frac{g\dot{x}}{u} \\ \ddot{y} &= -g \left(1 + \frac{\dot{y}}{u}\right)\end{aligned}$$

where g is the acceleration due to gravity.

You may assume without proof that the displacement equations are

$$\begin{aligned}x &= \frac{u^2}{g} \cos \theta \left(1 - e^{-\frac{gt}{u}}\right) \\ y &= \frac{u^2}{g} (1 + \sin \theta) \left(1 - e^{-\frac{gt}{u}}\right) - ut.\end{aligned}$$

- (i) Let the range of the motion be R . Show that **2**

$$R = \frac{u^2}{g} \cos \theta (1 - e^{-f(\theta)})$$

where $f(\theta) = \frac{gR}{u^2}(\sec \theta + \tan \theta)$.

- (ii) The student now wishes to find the launch angle such that the range of the object's motion is maximised. Let the range be $R(\theta)$ as a function of θ . Show that the launch angle where $R'(\theta) = 0$ satisfies **4**

$$\sin \theta = \frac{1}{e - 1}.$$

- (iii) Hence, show that R is maximised by a launch angle of $\theta = \sin^{-1} \left(\frac{1}{e - 1} \right)$. **2**

Question 16 continues on page 14

Question 16 (continued)

- (b) Suppose that, for the purposes of this question, either $\sqrt[3]{2}$ or $\sqrt[3]{4}$ is rational.
- (i) Prove the following statement. **2**
- $\sqrt[3]{2}$ is a rational number if and only if $\sqrt[3]{4}$ is a rational number.
- (ii) Let S_2 be the set of positive integers m such that $m\sqrt[3]{2}$ is an integer. **1**
- Similarly, let S_4 be the set of positive integers n such that $n\sqrt[3]{4}$ is an integer.
- Define $S = S_2 \cap S_4$. Show that there is at least one positive integer in S .
- (iii) Let k be any positive integer where $k \in S$. Show that $[(\sqrt[3]{4} - 1)k] \in S$. **2**
- (iv) Hence, prove that both $\sqrt[3]{2}$ and $\sqrt[3]{4}$ are, in fact, irrational. **2**

End of paper